RSPDBench: Benchmarking Vision Foundation Models on Earth Observation Tasks Under Physically Grounded Remote-Sensing Product Degradations

Supplementary Material

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This supplement gives the details referenced in the main paper. Section A describes the datasets and their splits. Section B lists every test condition, the band groups of the band-selective operators, and the noise scale of each run, with one example scene per dataset. Section C details the check of P1 and P3 against real Sentinel-2 bands. Section D reports the compound excess of every dataset–model pair, Sec. E the rank agreement between defect families, and Sec. F the degradation-aware readout. Section G lists the absolute clean and degraded scores behind every relative drop. The degradation operators, per-seed results, and analysis scripts are released at https://github.com/Tanjim131/RSPDBench.

A. Datasets

We use the five datasets as distributed by GEO-Bench [1] and PANGAEA [2], with their official splits (Tab. S1). For HLS Burn Scar, PANGAEA's training tiles are divided into training and validation sets (90/10, fixed random state), and its validation tiles are the test set. All degradations are applied to the products in their stored units, before any model-specific band selection, normalization, or resizing. Every model is selected on the clean validation set and evaluated on the same test images under every condition.

Table S1. Datasets. Bands are the spectral bands stored by the source; each model consumes the subset its input contract defines. †The m-SA-crop files also contain a cloud-probability layer, which no model uses. ‡Training and validation tiles come from PANGAEA's training split; the test tiles are its validation split.

Dataset	Source	Product	Bands	GSD	Size	Task	Metric	Train / val / test
m-EuroSAT	GEO-Bench	Sentinel-2 L1C	13	10 m	64^2	land cover (10 cl.)	Top-1 acc.	2,000 / 1,000 / 1,000
m-BigEarthNet	GEO-Bench	Sentinel-2 L2A	12	10 m	120^2	multi-label (43 cl.)	micro-F1	20,000 / 1,000 / 1,000
m-SA-crop	GEO-Bench	Sentinel-2	12†	10 m	256^2	crop type (10 cl.)	mIoU	3,000 / 1,000 / 1,000
m-Chesapeake	GEO-Bench	NAIP aerial	4	1 m	256^2	land cover (7 cl.)	mIoU	3,000 / 1,000 / 1,000
HLS Burn Scar	PANGAEA	HLS S30	6	30 m	512^2	burn scar (2 cl.)	mIoU	486 / 54 / 264‡

B. Degradation Conditions

B.1. Condition list

Table S2 lists the conditions. Models whose input contract exposes the multispectral band groups are evaluated on all 37 conditions; RGB and RGB+NIR contracts are evaluated on the 19 conditions without P3, C2, and the C2 arms. The two C2 arms apply one stage of C2 each and serve only as matched components for C2's compound excess (Sec. D). All random components are deterministic functions of the sample, the band, and the seed, and stay fixed across severities.

B.2. Band groups of P3 and C2

P3 shifts a group $\mathcal{S}$ of bands against fixed reference bands; C2 resamples and shifts a geometric group $\mathcal{G}$ and rescales a radiometric group $\mathcal{H}$ (Tab. S3). Each group is intersected with the bands a model consumes, so a model is affected only through bands it reads. B08 stays fixed under P3 on all Sentinel-2 datasets; C2 rescales it everywhere and, on m-BigEarthNet and m-SA-crop, also resamples and shifts it. The coarse bands B01, B09, and B10 are never in a group. GEO-Bench stores the m-EuroSAT bands in the physical order B01–B08, B09, B10, B11, B12, B8A under the nominal names B01–B08, B8A, B09–B12; every group refers to the physical bands.

Table S2. Test conditions. N: area-downsampling factor; α: gain magnitude; Δ: shift magnitude; k_s: noise fraction of the band mean μ_c. Chains use the settings of their components at the same severity. *Defined only for multispectral contracts.

Family	Operation	S1 / S2 / S3	Gain dir.	Conditions
P1	Gaussian PSF, $N \times N$ area downsampling, bilinear resize back	$\sigma = 0.5/1/2$ px, $N = 1/2/4$	–	3
P2	same gain on every band, $p = 1 \pm \alpha$	$\alpha = 3/8/15\%$	both	6
P3*	bilinear shift of group $\mathcal{S}$, fixed direction per sample	$\Delta = 0.3/1/2$ px	–	3
P4	Gaussian noise, std. $k_s \mu_c$	$k_s = 0.01/0.03/0.05$ (SNR $100/33/20$)	–	3
C1	P2 $\rightarrow$ P1 blur and downsampling $\rightarrow$ P4 on coarse grid $\rightarrow$ resize	matched severity	both	6
C2*	P1 and shift on $\mathcal{G}$, then gain on $\mathcal{H}$	matched severity	both	6
C2 res. arm*	P1 on $\mathcal{G}$ only (no shift, no gain)	matched severity	–	3
C2 gain arm*	gain on $\mathcal{H}$ only	matched severity	both	6
Total with clean: multispectral contracts / RGB or RGB+NIR contracts				37 / 19

Table S3. Band groups used by P3 and C2. Reference bands are never moved or rescaled.

Dataset	Reference	P3 shifted ($\mathcal{S}$)	C2 geometric ($\mathcal{G}$)	C2 radiometric ($\mathcal{H}$)
m-EuroSAT	B02–B04	B05–B07, B8A, B11, B12	B05–B07, B8A, B11, B12	B08, B8A, B11, B12
m-BigEarthNet	B02–B04	B05–B07, B8A, B11, B12	B08, B8A, B11, B12	B08, B8A, B11, B12
m-SA-crop	B02–B04	B05–B07, B8A, B11, B12	B08, B8A, B11, B12	B08, B8A, B11, B12
HLS Burn Scar	Blue, Green, Red	NIR, SWIR1, SWIR2	NIR, SWIR1, SWIR2	NIR, SWIR1, SWIR2
m-Chesapeake	RGB+NIR contract: P3 and C2 not defined			

B.3. Noise scale of P4 and C1

P4 and C1 scale the noise in band c by its clean mean μ_c (main paper, Sec. 3.1). Each evaluation pipeline obtained μ_c either by scanning the clean training split or from the dataset's published band statistics: GEO-Bench's band statistics or the PANGAEA dataset configuration. Table S4 compares the source used by every run with a full scan of the training split as loaded, $\bar{\mu}_c$. A different source changes the noise that is actually added, which has the effective signal-to-noise ratio

$$\mathrm{SNR}_{\mathrm{eff}} = \frac{\bar{\mu}_c}{k_s\, \mu_c} = \frac{1}{k_s}\, \frac{\bar{\mu}_c}{\mu_c}. \tag{1}$$

All sources agree with the training scan within about 3%, so the effective S3 SNR is about 19.5–20.6 for every run. On HLS Burn Scar, every model scales its noise by the training scan, taken over all 540 PANGAEA training tiles before the $90/10$ hold-out. The PANGAEA configuration means of this dataset differ from the distributed data by up to 39%; some models use them for input normalization, as their protocols prescribe, but no run uses them as μ_c.

B.4. Examples

Figure S1 shows one example scene per dataset under each family at S3, rendered with the released operators. Gain-signed families are shown in the gain-down direction. Each scene is displayed with a fixed false-color stretch taken from its clean image, so gain errors appear as brightness changes and misregistration as colored edges.

Table S4. Source of the noise scale μ_c per dataset and model, its largest deviation from the training-split mean $\bar{\mu}_c$ over the bands a model consumes, and the resulting effective SNR at S3 (nominal 20).

Dataset	Models	μ_c source	$\max_c \lvert\mu_c/\bar{\mu}_c - 1\rvert$	S3 $\mathrm{SNR}_{\mathrm{eff}}$
m-EuroSAT	DINOv3-Web, DINOv3-Sat, DOFA, Galileo, Panopticon, Prithvi 2.0, RAMEN	training scan	0.0%	20
m-EuroSAT	ViT-B/16	PANGAEA config	3.0%	20–21
m-BigEarthNet	DINOv3-Web, DINOv3-Sat, Panopticon	GEO-Bench stats	2.6%	20
m-BigEarthNet	DOFA, Galileo, Prithvi 2.0	training scan	0.0%	20
m-BigEarthNet	RAMEN, ViT-B/16	PANGAEA config	0.0%	20
m-SA-crop	DINOv3-Web, DINOv3-Sat, Galileo, Panopticon, Prithvi 2.0, U-Net	GEO-Bench stats	0.8%	20
m-SA-crop	DOFA, RAMEN	training scan	0.0%	20
m-Chesapeake	DINOv3-Web, DINOv3-Sat, Galileo, Panopticon, Prithvi 2.0, U-Net	GEO-Bench stats	1.2%	20
m-Chesapeake	DOFA, RAMEN	training scan	0.0%	20
HLS Burn Scar	DINOv3-Web, DINOv3-Sat, DOFA, Galileo, Panopticon, Prithvi 2.0, RAMEN, U-Net	training scan	0.0%	20

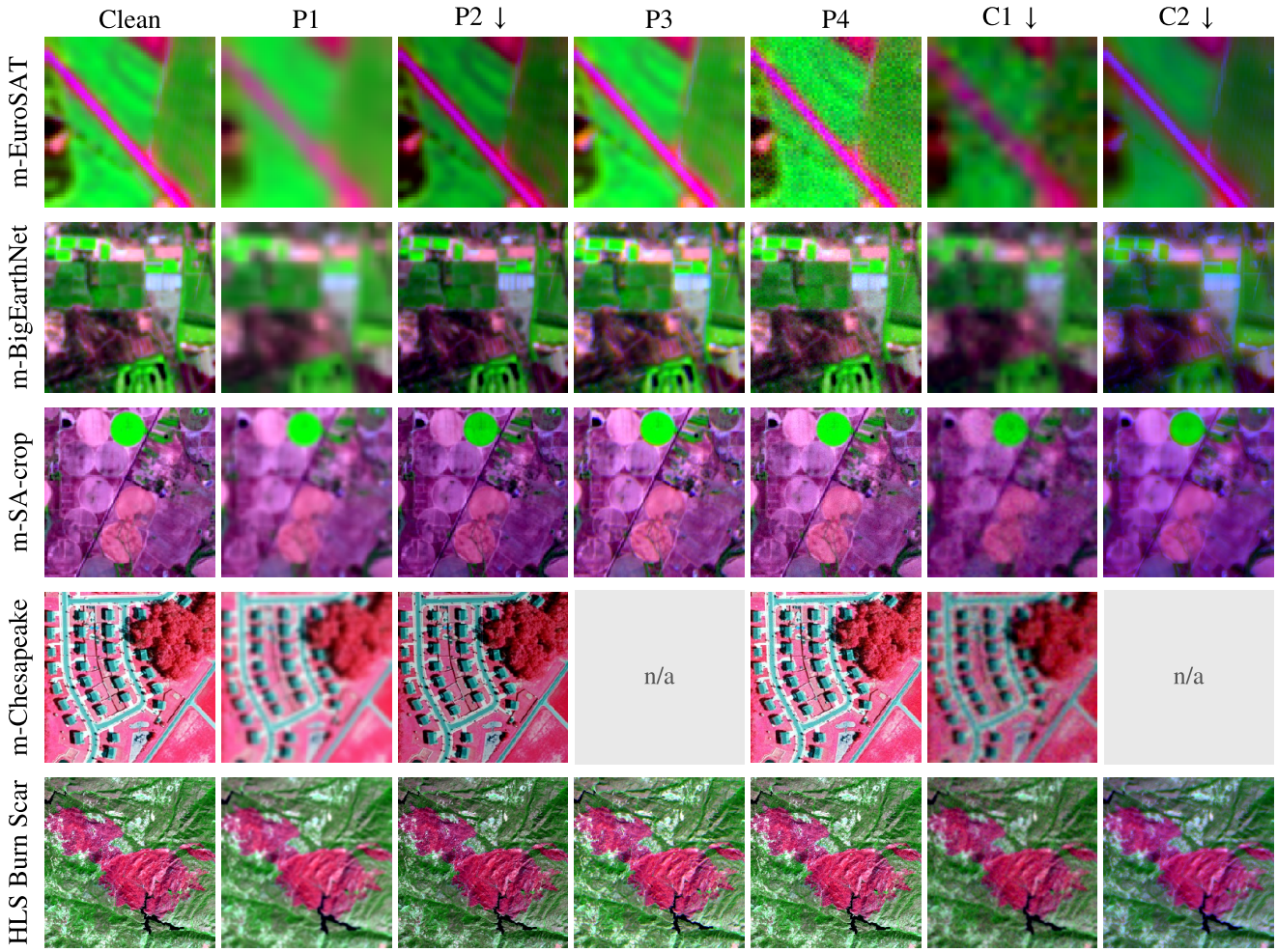


Figure S1. One example scene per dataset at S3. False color: SWIR 2, narrow NIR, and red for the Sentinel-2 datasets; SWIR 2, NIR, and red for HLS Burn Scar; NIR, red, and green for m-Chesapeake, where P3 and C2 are not defined.

C. Checking P1 and P3 Against Sentinel-2's Own Bands

Sentinel-2 records near-infrared light twice: band B08 at 10 m (833 nm) and band B8A at 20 m (865 nm). For each of the 1,000 test patches of the three Sentinel-2 datasets, we pass B08 through P1's Gaussian blur of width σ and 2×2 area

reduction, using the benchmark's own operator code, and correlate the result with the real B8A. Because the datasets store B8A upsampled to the 10 m grid (nearest-neighbor on m-SA-crop, smooth elsewhere), B8A is first area-averaged back to 20 m. On m-EuroSAT, B8A is read from its physical channel (Sec. B).

Figure S2 shows one patch and the median correlation as a function of σ. The best-fitting width brackets the $\sigma = N/2 = 1$ of P1-S2 on all three datasets, and removing the blur lowers the agreement (Tab. S5). We estimate the residual registration between B8A and P1-S2(B08) on the 10 m grid with Hann-windowed, amplitude-weighted cross-correlation at $1/50$ px resolution, on the half of the patches with the most texture (mean gradient magnitude relative to mean signal), because low-texture patches give unreliable estimates. The residual shift is 0.02–0.05 px (median), so P3-S1 (0.3 px) exceeds the residuals we observe in delivered products while remaining within the mission requirement, and P3-S2 and P3-S3 are deliberate stress levels.

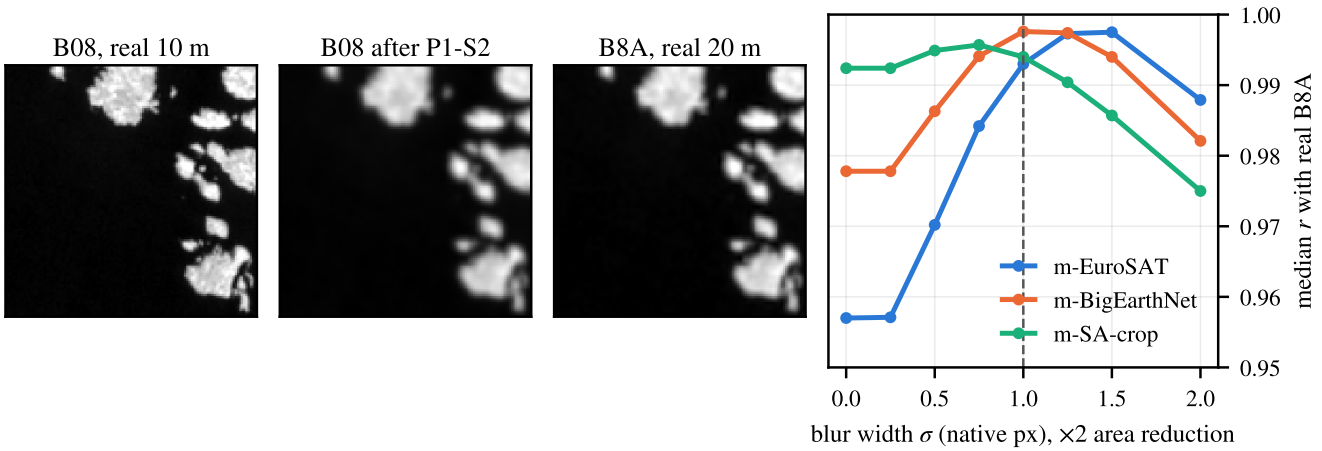


Figure S2. Left: one m-BigEarthNet test patch, near-infrared, same stretch for all three panels. Right: median Pearson correlation between the real 20 m B8A and the 10 m B08 after P1's blur of width σ and 2×2 area reduction; the dashed line marks the $\sigma = 1$ of P1-S2.

Table S5. P1 and P3 against real Sentinel-2 acquisitions. Left: median Pearson r between the real 20 m band and the blurred, area-reduced 10 m band; best σ is the median over patches of the best-fitting width (grid 0–2 in steps of 0.25). Right: residual B08–B8A displacement in native pixels on textured patches.

	Median r (20 m grid) at σ				Median	B08–B8A shift (px)	
Dataset	0	0.5	1.0	1.5	best σ	median	p95
m-EuroSAT	0.957	0.970	0.993	0.997	1.50	0.045	0.141
m-BigEarthNet	0.978	0.986	0.998	0.994	1.00	0.028	0.102
m-SA-crop	0.992	0.995	0.994	0.986	0.75	0.020	0.122

D. Compound Excess for Every Dataset and Model

A chain's excess is its relative drop minus the largest relative drop among its matched components at the same severity and gain direction, in percentage points, with both drops taken as means of the per-seed relative drops. For C1 the components are P1, P2, and P4; for C2 they are the resolution-only and gain-only arms (Tab. S2). C2 has no shift-only arm, so its excess can include the effect of the shift. Tables S6 and S7 report the largest excess over the three severities and both gain directions for every dataset–model pair, with the severity and direction where it occurs and the chain and component drops behind it. A positive value means the chain hurts more than its worst component alone; a value near zero means one component accounts for the whole chain.

Table S6. C1 excess (pp), maximum over severities and gain directions, as in main-paper Fig. 4. In parentheses: severity and gain direction, then the C1 drop and the largest component drop (%), both as means of the per-seed drops, so that their difference is the excess. Bold: largest value.

Model	m-EuroSAT	m-BigEarthNet	m-SA-crop	m-Chesapeake	HLS Burn Scar
DINOv3-Web	26.4 (S2↓; 44.0 vs 17.6)	12.1 (S2↓; 42.2 vs 30.2)	19.9 (S3↓; 75.4 vs 55.5)	16.0 (S3↓; 31.0 vs 15.0)	3.3 (S3↓; 21.2 vs 17.9)
DINOv3-Sat	3.8 (S1↓; 6.4 vs 2.7)	8.7 (S3↓; 98.0 vs 89.3)	**38.0** (S2↓; 61.7 vs 23.6)	28.3 (S3↓; 53.6 vs 25.3)	16.2 (S3↓; 30.1 vs 14.0)
DOFA	14.3 (S3↓; 49.7 vs 35.4)	3.9 (S3↓; 39.2 vs 35.3)	0.8 (S2↑; 23.6 vs 22.7)	29.7 (S3↓; 47.8 vs 18.0)	3.7 (S3↓; 12.8 vs 9.0)
Galileo	1.5 (S1↓; 5.6 vs 4.1)	7.3 (S3↓; 64.5 vs 57.2)	2.3 (S2↓; 15.2 vs 12.9)	4.3 (S2↓; 11.9 vs 7.5)	1.9 (S3↓; 2.8 vs 0.9)
Panopticon	15.6 (S3↓; 35.1 vs 19.5)	5.5 (S3↑; 48.8 vs 43.3)	27.0 (S3↓; 50.7 vs 23.7)	5.8 (S3↓; 14.0 vs 8.2)	18.6 (S3↓; 34.7 vs 16.1)
Prithvi 2.0	0.2 (S1↓; 2.8 vs 2.6)	2.3 (S3↓; 41.8 vs 39.5)	2.7 (S2↓; 33.2 vs 30.5)	13.7 (S3↓; 37.3 vs 23.6)	4.8 (S3↓; 7.7 vs 2.9)
RAMEN	8.0 (S3↓; 25.5 vs 17.5)	9.5 (S3↓; 25.3 vs 15.8)	7.2 (S3↓; 36.7 vs 29.5)	30.2 (S3↓; 40.4 vs 10.2)	-0.3 (S1↑; 0.5 vs 0.8)
ViT-B/16	10.5 (S3↓; 20.0 vs 9.5)	5.8 (S3↓; 15.4 vs 9.6)	–	–	–
U-Net	–	–	10.1 (S3↑; 65.3 vs 55.2)	15.9 (S3↓; 56.8 vs 40.9)	3.6 (S3↓; 13.1 vs 9.6)

Table S7. C2 excess (pp) over the larger of its two arms, for multispectral contracts. Layout as in Tab. S6.

Model	m-EuroSAT	m-BigEarthNet	m-SA-crop	HLS Burn Scar
DOFA	0.9 (S3↑; 1.8 vs 0.9)	3.2 (S3↓; 6.5 vs 3.3)	4.7 (S3↑; 11.3 vs 6.6)	1.2 (S3↓; 3.1 vs 1.9)
Galileo	**23.5** (S3↓; 36.0 vs 12.5)	17.3 (S3↓; 65.7 vs 48.4)	5.9 (S3↓; 43.8 vs 37.9)	0.4 (S3↑; 0.7 vs 0.3)
Panopticon	0.5 (S3↓; 1.0 vs 0.5)	9.5 (S3↓; 22.3 vs 12.8)	3.8 (S3↓; 11.7 vs 7.9)	-0.3 (S3↓; 1.2 vs 1.5)
Prithvi 2.0	2.9 (S3↓; 5.8 vs 2.9)	8.4 (S3↓; 24.3 vs 15.9)	11.7 (S3↑; 31.3 vs 19.6)	0.6 (S3↓; 2.0 vs 1.4)
RAMEN	9.1 (S3↓; 22.8 vs 13.7)	1.6 (S3↑; 11.5 vs 9.9)	2.0 (S3↑; 17.7 vs 15.7)	-0.2 (S1↓; -0.3 vs -0.0)
ViT-B/16	5.7 (S3↓; 10.0 vs 4.3)	6.7 (S3↑; 22.9 vs 16.3)	–	–
U-Net	–	–	13.2 (S2↑; 40.7 vs 27.5)	4.2 (S3↓; 9.7 vs 5.5)

E. Rank Agreement Across Defect Families

For each dataset, we rank the seven foundation models by their S3 drop under each family, using the worse gain direction for P2, C1, and C2, and compute Kendall's τ between every pair of rankings. Table S8 reports the median over datasets (5 datasets; 4 for pairs involving P3 or C2, which m-Chesapeake lacks). The last row gives the range over datasets of Spearman's ρ between clean score and S3 drop. It is positive on every dataset for P1 and negative on every dataset for P2, but changes sign across datasets for the other four families, so clean performance does not give one ordering across defect families.

Table S8. Median over datasets of Kendall's τ between the orderings of the foundation models by S3 relative drop under two families (7 models; 5 for P3 and C2). Last row: range over datasets of Spearman's ρ between clean score and S3 drop.

	P1	P2	P3	P4	C1	C2
P1	–	-0.52	+0.50	+0.43	+0.71	+0.30
P2	-0.52	–	+0.00	-0.33	-0.43	+0.30
P3	+0.50	+0.00	–	+0.60	+0.60	+0.10
P4	+0.43	-0.33	+0.60	–	+0.33	+0.30
C1	+0.71	-0.43	+0.60	+0.33	–	+0.20
C2	+0.30	+0.30	+0.10	+0.30	+0.20	–
ρ(clean, drop) min/max	+0.04/+0.79	-0.86/-0.32	-0.40/+0.70	-0.32/+0.57	-0.32/+0.96	-1.00/+0.70

F. Degradation-Aware Readout on m-BigEarthNet

To test which losses a better readout can recover, we retrain only the linear probe of each frozen encoder on degraded training views and keep everything else fixed. Each training image appears twice per epoch, and each view is degraded by a condition

drawn uniformly at random from those the model supports, clean included (37 for multispectral models, 19 for RGB); the probe's learning rate is selected on clean validation data. Relative drops are measured against the original clean-trained probe's clean micro-F1 and averaged over three seeds (Tab. S9). The readout sees the test defect types during training, so this is a supervised reference for defects that can be anticipated.

Table S9. Original clean-trained probes versus probes retrained on degraded views. "Clean change": relative change of the clean score; mean drop: over the model's P1–P4, C1 and C2 conditions, excluding clean and the C2 arms; right: S3 drop, original → degradation-aware.

	Clean	Mean drop, P1–P4, C1, C2		S3 drop: original → degradation-aware			
Model	change	orig.	aware	P1	P4	C1	C2
DINOv3-Web	-1.4	20.7	10.2	84.0→37.1	4.7→3.9	88.1→41.3	–
DINOv3-Sat	-2.0	29.7	12.3	89.3→40.2	17.4→7.0	98.0→48.5	–
DOFA	+1.0	7.3	2.5	35.3→16.8	1.3→-0.5	39.2→23.4	6.5→2.0
Galileo	-1.3	24.8	5.8	57.2→15.5	36.6→8.6	64.5→20.7	65.7→12.7
Panopticon	+1.0	12.0	2.2	43.3→10.5	20.9→3.7	48.8→15.5	22.3→6.0
Prithvi 2.0	+0.0	10.5	3.3	39.5→9.5	28.3→5.0	42.2→15.3	24.3→6.4
RAMEN	+1.7	7.3	1.9	15.8→8.2	0.4→-2.0	25.3→14.5	38.2→7.2

G. Absolute Clean and Degraded Scores

Tables S10 to S14 list the test score of every model under every condition: Top-1 accuracy on m-EuroSAT, micro-F1 at a fixed 0.5 threshold on m-BigEarthNet, and mIoU on the segmentation datasets, as mean ± standard deviation over three seeds, in %. The clean row (bold) is the reference for every relative drop in the main paper. Gain-signed conditions are listed per direction (↓ gain down, ↑ gain up); – marks conditions a model's input contract does not define.

Table S10. m-EuroSAT: Top-1 accuracy (%).

Condition	DINOv3-Web	DINOv3-Sat	DOFA	Galileo	Panopticon	Prithvi 2.0	RAMEN	ViT-B/16
Clean	**97.0** ±0.1	**95.3** ±0.1	**97.0** ±0.3	**97.2** ±0.1	**97.0** ±0.2	**93.9** ±0.1	**90.7** ±0.1	**92.7** ±0.4
P1 S1	96.9 ±0.1	94.2 ±0.2	96.4 ±0.5	94.2 ±0.6	97.1 ±0.2	91.4 ±0.6	89.9 ±0.2	92.6 ±0.3
P1 S2	79.9 ±0.4	44.0 ±1.1	84.7 ±2.1	40.0 ±1.3	90.0 ±0.6	73.8 ±0.4	85.2 ±0.4	91.9 ±0.2
P1 S3	37.3 ±1.0	18.3 ±0.3	62.7 ±2.5	31.2 ±2.0	78.1 ±0.2	54.9 ±1.0	75.8 ±0.1	86.9 ±1.6
P2↓ S1	97.0 ±0.1	95.2 ±0.1	96.9 ±0.3	97.2 ±0.1	97.1 ±0.2	93.5 ±0.3	90.2 ±0.1	92.2 ±0.2
P2↓ S2	96.8 ±0.0	95.0 ±0.1	96.6 ±0.2	97.2 ±0.1	97.2 ±0.1	90.7 ±0.3	85.9 ±0.3	90.6 ±0.3
P2↓ S3	96.5 ±0.1	94.8 ±0.2	95.8 ±0.1	96.9 ±0.2	96.6 ±0.2	82.8 ±0.1	74.8 ±0.3	83.9 ±1.6
P2↑ S1	97.1 ±0.1	95.2 ±0.2	97.2 ±0.2	97.2 ±0.1	97.0 ±0.2	93.8 ±0.1	90.7 ±0.1	92.3 ±0.5
P2↑ S2	96.8 ±0.1	95.3 ±0.1	97.1 ±0.4	97.2 ±0.1	97.0 ±0.2	92.8 ±0.3	88.8 ±0.2	91.3 ±0.3
P2↑ S3	96.8 ±0.1	95.3 ±0.2	96.3 ±0.3	97.0 ±0.2	96.5 ±0.3	89.2 ±0.6	83.1 ±0.1	87.4 ±0.7
P3 S1	–	–	97.1 ±0.3	97.2 ±0.1	97.0 ±0.2	94.1 ±0.1	90.8 ±0.2	92.5 ±0.4
P3 S2	–	–	97.0 ±0.2	96.7 ±0.2	97.1 ±0.1	94.0 ±0.3	91.0 ±0.1	92.3 ±0.6
P3 S3	–	–	96.8 ±0.2	94.3 ±1.1	97.2 ±0.2	93.4 ±0.3	90.6 ±0.2	91.2 ±0.5
P4 S1	97.0 ±0.1	92.7 ±0.2	96.9 ±0.3	93.2 ±0.5	97.2 ±0.2	93.4 ±0.4	90.7 ±0.2	92.7 ±0.6
P4 S2	94.6 ±0.1	70.7 ±0.6	96.0 ±0.1	79.4 ±2.3	93.5 ±0.2	85.9 ±0.3	87.4 ±0.1	92.5 ±0.4
P4 S3	90.1 ±0.4	58.4 ±0.8	93.7 ±0.5	65.4 ±4.9	85.0 ±0.5	69.7 ±0.7	76.8 ±0.2	92.6 ±0.4
C1↓ S1	95.7 ±0.4	89.1 ±0.4	96.1 ±0.3	91.7 ±0.2	97.1 ±0.1	91.2 ±0.4	89.7 ±0.1	91.9 ±0.3
C1↓ S2	54.3 ±0.2	53.8 ±0.5	77.0 ±1.3	47.7 ±8.6	87.2 ±0.2	74.8 ±0.8	82.1 ±0.3	88.8 ±1.2
C1↓ S3	23.1 ±0.1	26.4 ±1.2	48.8 ±1.0	32.1 ±4.8	63.0 ±1.1	60.0 ±1.0	67.5 ±0.2	74.2 ±2.7
C1↑ S1	96.1 ±0.2	89.7 ±0.3	96.3 ±0.3	92.4 ±0.1	96.9 ±0.3	91.8 ±0.6	89.7 ±0.1	92.2 ±0.5
C1↑ S2	62.2 ±0.3	55.9 ±0.0	82.6 ±1.1	51.5 ±7.1	88.6 ±0.3	75.6 ±0.1	82.0 ±0.1	90.1 ±1.1
C1↑ S3	25.0 ±0.6	28.4 ±1.5	63.3 ±1.4	38.8 ±4.6	72.0 ±0.6	62.1 ±0.8	68.8 ±0.1	83.0 ±0.1
C2↓ S1	–	–	97.0 ±0.1	97.2 ±0.2	97.1 ±0.2	94.0 ±0.2	90.4 ±0.3	92.7 ±0.5
C2↓ S2	–	–	96.8 ±0.1	91.8 ±1.8	96.7 ±0.3	93.0 ±0.2	87.3 ±0.1	91.1 ±0.8
C2↓ S3	–	–	96.0 ±0.1	62.2 ±6.3	96.0 ±0.1	88.5 ±0.2	70.0 ±0.1	83.4 ±0.5
C2↑ S1	–	–	97.0 ±0.2	97.2 ±0.1	97.0 ±0.2	94.1 ±0.2	90.4 ±0.2	92.5 ±0.2
C2↑ S2	–	–	96.6 ±0.2	95.3 ±1.0	97.0 ±0.3	92.9 ±0.0	88.8 ±0.1	91.3 ±0.3
C2↑ S3	–	–	95.2 ±0.7	86.0 ±2.3	95.9 ±0.0	88.5 ±0.2	79.7 ±0.2	88.2 ±0.9
C2 res. arm S1	–	–	97.0 ±0.3	97.2 ±0.1	97.1 ±0.2	94.0 ±0.1	90.7 ±0.1	92.7 ±0.4
C2 res. arm S2	–	–	96.9 ±0.3	96.7 ±0.2	97.4 ±0.1	93.6 ±0.2	90.8 ±0.2	92.4 ±0.3
C2 res. arm S3	–	–	96.3 ±0.4	90.9 ±1.9	97.0 ±0.2	91.2 ±0.5	89.2 ±0.1	90.7 ±0.4
C2 gain arm↓ S1	–	–	97.0 ±0.1	97.1 ±0.2	97.2 ±0.2	93.9 ±0.1	90.6 ±0.4	92.7 ±0.5
C2 gain arm↓ S2	–	–	96.8 ±0.2	95.5 ±0.8	96.9 ±0.1	93.6 ±0.2	89.4 ±0.2	91.8 ±0.4
C2 gain arm↓ S3	–	–	96.8 ±0.3	85.1 ±5.8	96.5 ±0.1	91.6 ±0.5	78.2 ±0.2	88.7 ±0.8
C2 gain arm↑ S1	–	–	97.1 ±0.3	97.2 ±0.0	96.8 ±0.0	93.9 ±0.2	90.4 ±0.2	92.4 ±0.3
C2 gain arm↑ S2	–	–	96.8 ±0.4	96.6 ±0.1	96.7 ±0.1	93.0 ±0.3	87.6 ±0.1	91.7 ±0.6
C2 gain arm↑ S3	–	–	96.1 ±0.4	94.5 ±0.6	96.3 ±0.2	92.0 ±0.2	77.7 ±0.4	88.8 ±1.0

Table S11. m-BigEarthNet: micro-F1 (%).

Condition	DINOv3-Web	DINOv3-Sat	DOFA	Galileo	Panopticon	Prithvi 2.0	RAMEN	ViT-B/16
Clean	**64.0** ±0.1	**63.5** ±0.1	**58.0** ±0.3	**62.8** ±0.3	**66.6** ±0.5	**55.6** ±0.7	**52.9** ±0.7	**64.9** ±0.4
P1 S1	63.4 ±0.2	63.3 ±0.1	56.8 ±0.2	60.0 ±0.2	64.5 ±0.7	55.3 ±1.1	52.5 ±0.7	65.2 ±0.5
P1 S2	44.7 ±0.4	17.4 ±0.9	47.9 ±0.2	30.1 ±0.2	47.9 ±1.4	42.7 ±0.9	48.1 ±0.5	63.7 ±0.7
P1 S3	10.2 ±0.1	6.8 ±0.9	37.5 ±0.0	26.9 ±0.2	37.7 ±1.1	33.6 ±0.8	44.5 ±0.3	58.7 ±0.6
P2↓ S1	63.9 ±0.1	63.6 ±0.1	57.9 ±0.2	62.5 ±0.3	66.5 ±0.5	55.9 ±0.8	52.6 ±0.5	64.8 ±0.5
P2↓ S2	63.8 ±0.2	63.4 ±0.2	57.6 ±0.2	61.9 ±0.3	66.3 ±0.6	56.1 ±0.7	51.0 ±0.7	64.2 ±0.5
P2↓ S3	63.8 ±0.2	63.0 ±0.2	56.7 ±0.1	60.2 ±0.4	65.2 ±0.7	56.1 ±0.8	47.2 ±1.2	61.7 ±0.7
P2↑ S1	64.1 ±0.1	63.5 ±0.1	58.0 ±0.1	63.1 ±0.4	66.3 ±0.4	55.4 ±0.7	53.1 ±0.7	64.8 ±0.7
P2↑ S2	64.2 ±0.1	63.8 ±0.2	58.1 ±0.2	62.9 ±0.3	66.1 ±0.6	54.7 ±0.6	52.6 ±0.6	64.4 ±0.5
P2↑ S3	64.2 ±0.1	63.9 ±0.0	57.7 ±0.3	62.1 ±0.1	64.9 ±0.6	53.6 ±0.6	50.7 ±0.5	63.1 ±0.3
P3 S1	–	–	57.8 ±0.2	62.7 ±0.2	66.4 ±0.5	56.0 ±0.5	52.8 ±0.7	64.9 ±0.4
P3 S2	–	–	57.5 ±0.3	60.7 ±0.2	66.3 ±0.7	55.5 ±0.5	52.6 ±0.5	65.0 ±0.4
P3 S3	–	–	57.0 ±0.4	60.7 ±0.1	65.1 ±0.7	53.2 ±0.3	52.0 ±0.4	63.6 ±0.3
P4 S1	63.9 ±0.1	63.4 ±0.0	58.0 ±0.2	59.5 ±0.5	66.2 ±0.5	54.6 ±0.9	53.0 ±0.6	64.9 ±0.4
P4 S2	62.8 ±0.1	60.6 ±0.2	57.7 ±0.2	49.6 ±0.8	61.0 ±1.0	49.0 ±0.6	52.9 ±0.6	65.0 ±0.5
P4 S3	61.0 ±0.1	52.5 ±0.2	57.3 ±0.4	39.8 ±0.7	52.7 ±1.4	39.9 ±1.7	52.7 ±0.6	65.1 ±0.5
C1↓ S1	63.6 ±0.1	63.3 ±0.2	56.6 ±0.0	57.2 ±0.3	64.6 ±0.6	54.5 ±1.3	51.9 ±0.7	65.1 ±0.6
C1↓ S2	37.0 ±0.1	14.1 ±0.4	47.4 ±0.2	30.2 ±0.2	48.7 ±1.3	45.3 ±0.3	46.4 ±0.4	62.4 ±0.3
C1↓ S3	7.6 ±0.1	1.3 ±0.1	35.2 ±0.0	22.3 ±0.1	35.5 ±1.5	32.4 ±1.2	39.5 ±1.0	54.9 ±0.7
C1↑ S1	63.5 ±0.1	63.5 ±0.1	56.9 ±0.2	57.9 ±0.2	64.3 ±0.5	54.3 ±1.2	52.4 ±0.8	64.9 ±0.5
C1↑ S2	40.9 ±0.1	14.3 ±0.6	47.5 ±0.1	31.0 ±0.1	47.7 ±1.2	45.0 ±0.5	47.7 ±0.1	63.8 ±0.2
C1↑ S3	11.3 ±0.2	2.3 ±0.5	36.5 ±0.2	22.8 ±0.1	34.1 ±1.5	32.5 ±0.9	42.5 ±0.4	56.7 ±0.6
C2↓ S1	–	–	57.6 ±0.1	54.5 ±0.3	66.0 ±1.0	56.8 ±0.5	49.7 ±1.3	64.2 ±0.5
C2↓ S2	–	–	56.6 ±0.2	32.2 ±0.5	62.0 ±0.6	55.0 ±0.1	42.0 ±0.7	59.8 ±0.5
C2↓ S3	–	–	54.2 ±0.6	21.6 ±0.2	51.7 ±1.0	42.1 ±1.1	32.7 ±0.7	48.8 ±1.1
C2↑ S1	–	–	57.9 ±0.2	54.8 ±0.3	66.2 ±0.8	56.4 ±0.5	53.5 ±0.0	64.7 ±0.4
C2↑ S2	–	–	57.0 ±0.2	37.9 ±0.3	63.2 ±0.6	54.9 ±0.4	51.6 ±0.3	61.1 ±0.3
C2↑ S3	–	–	55.5 ±0.3	30.4 ±0.4	53.6 ±0.7	43.9 ±0.4	46.9 ±0.5	50.0 ±1.2
C2 res. arm S1	–	–	57.9 ±0.1	58.9 ±0.1	66.3 ±0.5	56.4 ±0.7	52.9 ±0.6	65.0 ±0.4
C2 res. arm S2	–	–	57.3 ±0.1	38.2 ±0.3	64.5 ±0.6	56.5 ±0.1	51.7 ±0.6	64.3 ±0.3
C2 res. arm S3	–	–	56.1 ±0.2	32.4 ±0.5	58.0 ±0.4	46.8 ±0.7	50.0 ±0.4	61.3 ±0.4
C2 gain arm↓ S1	–	–	57.8 ±0.2	62.1 ±0.2	66.5 ±0.6	55.8 ±0.7	50.0 ±1.2	64.3 ±0.8
C2 gain arm↓ S2	–	–	57.5 ±0.2	58.5 ±0.3	65.5 ±0.5	55.7 ±0.7	42.7 ±0.9	61.0 ±0.5
C2 gain arm↓ S3	–	–	56.6 ±0.3	47.5 ±0.2	61.8 ±0.7	55.0 ±0.8	33.0 ±0.9	52.8 ±0.7
C2 gain arm↑ S1	–	–	58.0 ±0.2	62.4 ±0.3	66.3 ±0.4	55.4 ±0.7	53.7 ±0.1	64.6 ±0.4
C2 gain arm↑ S2	–	–	57.9 ±0.2	59.8 ±0.1	65.4 ±0.4	54.7 ±0.7	51.9 ±0.3	61.8 ±0.4
C2 gain arm↑ S3	–	–	57.4 ±0.5	54.1 ±0.3	63.3 ±0.5	53.5 ±0.9	47.7 ±0.3	54.4 ±0.9

Table S12. m-SA-crop: mIoU (%).

Condition	DINOv3-Web	DINOv3-Sat	DOFA	Galileo	Panopticon	Prithvi 2.0	RAMEN	U-Net
Clean	**31.8** ±0.1	**32.9** ±0.1	**28.3** ±0.3	**16.2** ±0.0	**27.9** ±0.4	**26.1** ±0.1	**18.7** ±0.0	**34.9** ±0.4
P1 S1	31.7 ±0.2	32.6 ±0.3	28.7 ±0.2	16.2 ±0.0	27.7 ±0.1	25.6 ±0.0	18.8 ±0.0	34.4 ±0.3
P1 S2	24.6 ±0.4	25.1 ±0.4	21.9 ±0.1	14.2 ±0.0	26.7 ±0.1	18.1 ±0.2	19.0 ±0.0	20.8 ±0.6
P1 S3	14.1 ±0.0	7.8 ±0.4	11.2 ±0.2	9.0 ±0.1	21.3 ±0.5	12.7 ±0.1	19.1 ±0.0	15.6 ±0.6
P2↓ S1	31.7 ±0.2	32.8 ±0.1	28.1 ±0.3	15.9 ±0.0	27.9 ±0.4	25.6 ±0.1	18.0 ±0.0	34.6 ±0.4
P2↓ S2	31.5 ±0.2	32.7 ±0.1	27.5 ±0.3	15.2 ±0.0	27.7 ±0.5	24.5 ±0.1	16.4 ±0.0	33.8 ±0.5
P2↓ S3	31.4 ±0.2	32.6 ±0.1	26.4 ±0.4	14.3 ±0.0	27.4 ±0.6	22.3 ±0.1	13.2 ±0.0	31.9 ±0.7
P2↑ S1	31.8 ±0.2	32.9 ±0.1	28.5 ±0.3	16.6 ±0.0	28.0 ±0.3	26.5 ±0.1	19.1 ±0.0	35.0 ±0.4
P2↑ S2	31.8 ±0.2	33.0 ±0.1	28.7 ±0.2	17.2 ±0.0	28.0 ±0.3	26.9 ±0.1	19.3 ±0.0	34.8 ±0.4
P2↑ S3	31.9 ±0.2	33.0 ±0.1	28.6 ±0.2	17.8 ±0.0	28.0 ±0.2	27.3 ±0.1	18.8 ±0.0	33.7 ±0.5
P3 S1	–	–	28.4 ±0.3	16.3 ±0.0	27.6 ±0.1	26.0 ±0.1	18.7 ±0.0	34.9 ±0.3
P3 S2	–	–	28.3 ±0.3	16.3 ±0.0	27.4 ±0.1	24.9 ±0.1	18.5 ±0.0	34.2 ±0.2
P3 S3	–	–	27.4 ±0.3	16.2 ±0.0	27.2 ±0.1	23.3 ±0.2	18.4 ±0.0	32.5 ±0.2
P4 S1	31.6 ±0.1	32.6 ±0.1	28.3 ±0.3	16.2 ±0.0	28.0 ±0.4	25.9 ±0.1	18.7 ±0.0	34.9 ±0.4
P4 S2	29.6 ±0.4	28.3 ±0.4	28.2 ±0.3	15.8 ±0.0	27.4 ±0.7	24.3 ±0.2	18.4 ±0.0	33.6 ±0.8
P4 S3	26.4 ±0.9	20.0 ±0.8	27.9 ±0.3	14.6 ±0.0	24.9 ±0.9	22.6 ±0.3	18.0 ±0.0	28.1 ±1.6
C1↓ S1	31.5 ±0.1	32.4 ±0.3	28.5 ±0.2	15.8 ±0.0	27.8 ±0.2	25.0 ±0.0	18.1 ±0.0	34.2 ±0.3
C1↓ S2	18.8 ±0.8	12.6 ±0.5	22.5 ±0.0	13.8 ±0.0	26.6 ±0.6	17.4 ±0.1	16.4 ±0.0	21.3 ±0.6
C1↓ S3	7.8 ±0.2	6.0 ±0.1	13.5 ±0.2	11.2 ±0.0	13.8 ±1.7	14.0 ±0.2	11.8 ±0.0	14.7 ±1.0
C1↑ S1	31.6 ±0.1	32.5 ±0.3	28.8 ±0.2	16.5 ±0.0	27.8 ±0.1	25.5 ±0.0	19.1 ±0.0	34.5 ±0.2
C1↑ S2	20.4 ±0.9	14.5 ±0.6	21.7 ±0.1	15.3 ±0.1	27.3 ±0.2	17.5 ±0.1	19.2 ±0.0	20.3 ±0.5
C1↑ S3	8.7 ±0.3	5.7 ±0.1	12.2 ±0.2	13.6 ±0.0	18.1 ±1.5	12.5 ±0.3	17.9 ±0.0	12.1 ±0.9
C2↓ S1	–	–	28.1 ±0.3	15.9 ±0.0	27.2 ±0.1	25.1 ±0.2	17.4 ±0.0	31.4 ±0.6
C2↓ S2	–	–	27.4 ±0.3	13.7 ±0.0	26.3 ±0.1	22.1 ±0.2	15.0 ±0.0	19.3 ±1.2
C2↓ S3	–	–	25.5 ±0.3	9.1 ±0.1	24.7 ±0.2	18.8 ±0.3	12.1 ±0.0	8.6 ±0.9
C2↑ S1	–	–	28.5 ±0.2	16.4 ±0.0	27.7 ±0.0	25.9 ±0.1	19.3 ±0.0	33.5 ±0.4
C2↑ S2	–	–	27.9 ±0.1	15.1 ±0.0	27.2 ±0.3	23.0 ±0.2	18.4 ±0.0	20.7 ±0.9
C2↑ S3	–	–	25.1 ±0.2	10.9 ±0.1	25.7 ±0.9	17.9 ±0.3	15.4 ±0.0	12.6 ±1.0
C2 res. arm S1	–	–	28.4 ±0.3	16.2 ±0.0	27.7 ±0.2	26.0 ±0.1	18.7 ±0.0	34.8 ±0.3
C2 res. arm S2	–	–	28.5 ±0.3	14.6 ±0.0	27.2 ±0.1	23.9 ±0.1	18.5 ±0.0	29.6 ±0.4
C2 res. arm S3	–	–	28.5 ±0.2	10.1 ±0.1	26.6 ±0.5	21.0 ±0.2	18.3 ±0.0	22.7 ±1.2
C2 gain arm↓ S1	–	–	28.0 ±0.3	16.0 ±0.0	27.6 ±0.4	25.2 ±0.1	17.5 ±0.0	32.1 ±0.7
C2 gain arm↓ S2	–	–	27.1 ±0.3	15.5 ±0.0	26.9 ±0.4	23.3 ±0.1	15.2 ±0.0	22.6 ±1.4
C2 gain arm↓ S3	–	–	25.2 ±0.3	14.8 ±0.0	25.7 ±0.4	20.3 ±0.0	12.4 ±0.0	9.3 ±1.0
C2 gain arm↑ S1	–	–	28.5 ±0.2	16.5 ±0.0	28.1 ±0.3	26.7 ±0.1	19.3 ±0.0	34.3 ±0.3
C2 gain arm↑ S2	–	–	28.1 ±0.1	16.9 ±0.0	28.3 ±0.3	27.3 ±0.1	18.7 ±0.0	25.3 ±2.1
C2 gain arm↑ S3	–	–	26.5 ±0.0	17.4 ±0.0	28.1 ±0.2	27.0 ±0.1	15.8 ±0.0	14.2 ±2.9

Table S13. m-Chesapeake: mIoU (%).

Condition	DINOv3-Web	DINOv3-Sat	DOFA	Galileo	Panopticon	Prithvi 2.0	RAMEN	U-Net
Clean	**61.7** ±0.9	**62.3** ±5.6	**52.8** ±0.1	**50.5** ±0.6	**65.1** ±0.5	**64.4** ±0.4	**44.3** ±0.0	**71.8** ±0.3
P1 S1	61.7 ±1.0	62.6 ±5.4	52.4 ±0.1	50.5 ±0.5	65.3 ±0.6	63.9 ±0.4	44.5 ±0.0	71.5 ±0.4
P1 S2	58.0 ±1.3	57.2 ±5.8	49.7 ±0.1	46.7 ±0.3	62.8 ±0.6	59.3 ±0.5	43.9 ±0.0	62.9 ±1.5
P1 S3	52.4 ±0.5	46.6 ±5.9	43.3 ±0.1	32.4 ±0.3	59.8 ±0.2	49.2 ±0.2	41.4 ±0.0	42.5 ±0.9
P2↓ S1	61.7 ±1.0	62.3 ±5.6	52.7 ±0.1	50.6 ±0.6	65.2 ±0.5	64.1 ±0.4	44.3 ±0.0	71.8 ±0.3
P2↓ S2	61.7 ±1.0	62.1 ±5.7	52.0 ±0.1	50.1 ±0.6	65.2 ±0.5	63.0 ±0.4	44.2 ±0.0	71.2 ±0.2
P2↓ S3	61.7 ±1.0	62.0 ±5.7	50.3 ±0.2	49.1 ±0.6	65.2 ±0.5	57.4 ±0.5	43.8 ±0.0	69.5 ±0.5
P2↑ S1	61.7 ±0.9	62.4 ±5.6	52.8 ±0.2	50.1 ±0.6	65.1 ±0.5	64.5 ±0.4	44.4 ±0.0	71.7 ±0.3
P2↑ S2	61.7 ±0.9	62.4 ±5.5	52.6 ±0.2	49.3 ±0.6	65.1 ±0.5	64.0 ±0.3	44.4 ±0.0	71.2 ±0.3
P2↑ S3	61.5 ±0.9	62.2 ±5.5	51.9 ±0.3	47.9 ±0.6	65.0 ±0.5	62.6 ±0.2	44.3 ±0.0	69.7 ±0.4
P4 S1	57.8 ±0.1	57.6 ±0.1	52.8 ±0.1	50.5 ±0.5	65.1 ±0.5	64.4 ±0.4	44.2 ±0.0	71.8 ±0.4
P4 S2	56.6 ±0.1	56.2 ±0.1	52.6 ±0.1	50.1 ±0.5	64.2 ±0.5	64.4 ±0.5	43.0 ±0.0	71.2 ±0.6
P4 S3	55.4 ±0.1	54.2 ±0.2	52.1 ±0.1	49.4 ±0.5	60.2 ±0.8	63.9 ±0.6	39.8 ±0.0	70.1 ±0.9
C1↓ S1	57.7 ±0.1	57.5 ±0.1	52.3 ±0.0	50.4 ±0.5	65.3 ±0.5	63.4 ±0.4	44.4 ±0.0	71.2 ±0.4
C1↓ S2	54.5 ±0.4	49.9 ±0.3	47.7 ±0.3	44.5 ±0.3	62.3 ±0.8	53.1 ±0.8	43.2 ±0.0	59.6 ±1.3
C1↓ S3	42.6 ±0.5	28.8 ±0.2	27.6 ±0.3	33.5 ±0.4	56.0 ±1.6	40.4 ±0.9	26.4 ±0.1	31.0 ±3.8
C1↑ S1	57.8 ±0.1	57.6 ±0.1	52.5 ±0.1	50.1 ±0.5	65.3 ±0.5	64.2 ±0.4	44.5 ±0.0	71.4 ±0.5
C1↑ S2	55.1 ±0.4	50.9 ±0.5	48.9 ±0.1	46.6 ±0.3	62.5 ±0.8	60.6 ±0.5	43.7 ±0.0	62.6 ±0.8
C1↑ S3	45.3 ±0.6	35.3 ±1.0	34.8 ±0.4	33.6 ±0.2	57.4 ±0.8	52.1 ±0.3	37.4 ±0.1	41.1 ±1.1

Table S14. HLS Burn Scar: mIoU (%).

Condition	DINOv3-Web	DINOv3-Sat	DOFA	Galileo	Panopticon	Prithvi 2.0	RAMEN	U-Net
Clean	**80.6** ±0.1	**80.6** ±0.1	**81.3** ±0.2	**81.6** ±0.1	**75.9** ±0.5	**80.0** ±1.2	**70.0** ±0.1	**84.9** ±0.5
P1 S1	80.8 ±0.2	80.2 ±0.1	81.1 ±0.2	81.6 ±0.1	76.3 ±0.7	80.0 ±1.2	70.2 ±0.1	84.7 ±0.5
P1 S2	79.1 ±0.3	75.8 ±0.6	78.5 ±0.2	81.3 ±0.1	74.2 ±0.8	79.5 ±1.4	71.0 ±0.1	81.8 ±0.7
P1 S3	66.2 ±0.3	69.3 ±0.6	74.0 ±0.4	80.9 ±0.1	63.7 ±7.8	77.7 ±1.6	71.8 ±0.1	76.8 ±0.6
P2↓ S1	80.6 ±0.1	80.6 ±0.1	81.1 ±0.2	81.6 ±0.1	76.0 ±0.6	79.9 ±1.2	70.5 ±0.1	84.9 ±0.5
P2↓ S2	80.5 ±0.1	80.6 ±0.1	80.8 ±0.2	81.5 ±0.1	75.9 ±0.7	79.5 ±1.2	70.9 ±0.1	84.9 ±0.6
P2↓ S3	80.4 ±0.2	80.5 ±0.1	80.0 ±0.2	80.9 ±0.1	75.7 ±0.8	78.6 ±1.3	70.4 ±0.1	84.8 ±0.6
P2↑ S1	80.7 ±0.1	80.5 ±0.1	81.5 ±0.2	81.6 ±0.1	75.9 ±0.5	80.1 ±1.2	69.5 ±0.1	84.9 ±0.5
P2↑ S2	80.7 ±0.1	80.5 ±0.1	81.7 ±0.1	81.5 ±0.1	75.8 ±0.4	80.1 ±1.2	68.5 ±0.1	84.8 ±0.6
P2↑ S3	80.7 ±0.1	80.4 ±0.2	81.8 ±0.2	81.3 ±0.1	75.6 ±0.5	80.0 ±1.2	67.0 ±0.2	84.5 ±0.8
P3 S1	–	–	81.2 ±0.3	81.6 ±0.1	76.2 ±0.6	80.0 ±1.2	70.1 ±0.1	84.8 ±0.5
P3 S2	–	–	80.8 ±0.5	81.6 ±0.0	76.2 ±0.7	79.9 ±1.2	69.9 ±0.1	84.3 ±0.4
P3 S3	–	–	80.1 ±0.7	81.5 ±0.1	75.5 ±0.7	79.6 ±1.2	69.5 ±0.1	83.1 ±0.4
P4 S1	80.6 ±0.2	80.5 ±0.1	81.3 ±0.2	81.6 ±0.1	75.9 ±0.5	80.0 ±1.2	70.0 ±0.1	84.9 ±0.5
P4 S2	79.9 ±0.3	80.0 ±0.1	81.2 ±0.2	81.6 ±0.1	75.8 ±0.4	79.9 ±1.2	69.8 ±0.2	84.9 ±0.5
P4 S3	78.9 ±0.4	79.0 ±0.1	81.0 ±0.2	81.5 ±0.1	75.5 ±0.4	79.8 ±1.2	69.4 ±0.2	84.7 ±0.5
C1↓ S1	80.6 ±0.2	80.0 ±0.1	80.9 ±0.2	81.5 ±0.1	76.2 ±0.7	79.8 ±1.2	70.6 ±0.1	84.7 ±0.4
C1↓ S2	76.8 ±0.2	73.6 ±1.1	77.1 ±0.2	81.3 ±0.1	69.9 ±1.4	78.5 ±1.5	71.3 ±0.1	81.9 ±0.6
C1↓ S3	63.5 ±2.0	56.3 ±2.0	70.9 ±0.4	79.3 ±0.3	49.6 ±6.5	73.9 ±2.1	70.1 ±0.1	73.7 ±2.1
C1↑ S1	80.7 ±0.2	80.2 ±0.1	81.2 ±0.2	81.5 ±0.1	76.3 ±0.6	80.0 ±1.2	69.7 ±0.1	84.7 ±0.5
C1↑ S2	77.3 ±0.3	73.6 ±1.2	79.1 ±0.3	81.2 ±0.1	71.2 ±1.5	79.8 ±1.3	69.2 ±0.1	82.4 ±1.0
C1↑ S3	66.6 ±1.2	57.0 ±1.9	73.6 ±0.6	80.3 ±0.1	51.2 ±7.9	78.3 ±1.3	67.9 ±0.2	78.5 ±0.7
C2↓ S1	–	–	81.0 ±0.3	81.4 ±0.0	76.2 ±0.7	79.9 ±1.2	70.2 ±0.1	84.1 ±0.6
C2↓ S2	–	–	80.1 ±0.4	80.9 ±0.1	75.8 ±1.2	79.5 ±1.1	70.3 ±0.1	81.4 ±0.8
C2↓ S3	–	–	78.8 ±0.4	80.1 ±0.1	75.0 ±1.0	78.4 ±1.1	70.0 ±0.1	76.6 ±0.8
C2↑ S1	–	–	81.3 ±0.2	81.7 ±0.1	76.6 ±0.8	80.0 ±1.2	70.2 ±0.1	84.9 ±0.4
C2↑ S2	–	–	80.7 ±0.4	81.6 ±0.1	76.9 ±1.1	79.8 ±1.2	70.3 ±0.2	83.9 ±0.5
C2↑ S3	–	–	79.9 ±0.4	81.0 ±0.2	77.0 ±1.1	79.0 ±1.3	70.4 ±0.2	79.4 ±0.5
C2 res. arm S1	–	–	81.3 ±0.2	81.6 ±0.1	76.2 ±0.7	80.0 ±1.2	70.1 ±0.1	84.8 ±0.5
C2 res. arm S2	–	–	80.8 ±0.1	81.5 ±0.1	76.6 ±1.2	79.7 ±1.2	70.5 ±0.1	83.4 ±0.6
C2 res. arm S3	–	–	79.8 ±0.1	81.4 ±0.1	76.6 ±1.4	78.9 ±1.2	70.7 ±0.1	80.2 ±0.5
C2 gain arm↓ S1	–	–	81.2 ±0.2	81.4 ±0.1	75.7 ±0.5	79.9 ±1.2	70.0 ±0.1	84.5 ±0.7
C2 gain arm↓ S2	–	–	80.8 ±0.2	81.0 ±0.1	75.4 ±0.5	79.8 ±1.2	70.0 ±0.1	83.5 ±0.8
C2 gain arm↓ S3	–	–	80.1 ±0.2	80.3 ±0.2	74.8 ±0.6	79.5 ±1.1	69.6 ±0.1	81.6 ±1.1
C2 gain arm↑ S1	–	–	81.4 ±0.2	81.7 ±0.1	76.1 ±0.5	80.1 ±1.2	70.0 ±0.1	85.2 ±0.4
C2 gain arm↑ S2	–	–	81.6 ±0.2	81.8 ±0.1	76.3 ±0.5	80.2 ±1.2	69.9 ±0.2	85.6 ±0.4
C2 gain arm↑ S3	–	–	81.5 ±0.2	81.6 ±0.2	76.6 ±0.6	80.2 ±1.1	69.8 ±0.2	85.5 ±0.4

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